



Xi'an Jiaotong-Liverpool University

西交利物浦大學

FIN302

Finance and Markets

Lectures 7-12: Revision

Prof. Tony He

An Overview from the First Lecture

➤ Contents

- Derivative markets and futures hedging (Lect. 7-8)
- Swaps (Lect. 9)
- Options, pricing, and hedging (Lect. 10-12)

➤ Economic Intuitions are your starting point!

- Trade-off between benefits and costs; a better return for high-risk investment
- You pay what you get in the future: expected payoff and discounting

➤ Economic mechanisms are crucial for a good understanding

- Pricing: no-arbitrage principle or no free lunch
 - If your investment has no risk, your return should be given by a risk-free rate---option hedging and pricing
 - If you invest nothing, you will get nothing; e.g., your return should be zero

Revision---Main Questions, Measures, and Approaches

- What are the risks?
 - What are the derivatives and how do we use them?
 - What are the specific risks for different derivatives: futures, swaps, options
- How to measure the risk?
 - Equity portfolio: market risk and CAPM
 - Option risks: Greek letters---delta, gamma, vega
- How to hedge the risks?
 - Forward, swaps, options---binominal trees and BS
 - Equity portfolio risks: index futures and CAPM
 - Hedge interest rate and currency risks: swaps
 - Hedge option portfolio risks: delta, gamma, vega neutral
 - Portfolio insurance via index put options



Derivatives

A **derivative** is a financial instrument

- Involves two parties agreeing to future transactions
- whose value depends on the value(s) of other underlying variables.

Derivatives:

- Forward/futures
- Options
- Swaps
- Interest rate futures/options
- Credit...electricity...weather...insurance
- ...

Derivatives

What are the underlying assets:

- Stocks, indices, and futures
- Exchange rate
- Interest rate
- Any thing...hogs...snow falling...

Who care:

- Investors,
- Traders/dealers
- Financial intuitions
- Fund and risk managers
- Firms/corporates
- Market regulators/policy makers
- ...



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Derivative Markets: what for

Derivatives are Used

- To hedge/manage risks
- To speculate (by taking a view on the future direction of the market)
- To lock in an arbitrage profit
- To change the nature of
 - a liability
 - an investment

Aims of the Module

- The aim of this module related to the 2nd part is to give students a ***solid grounding of the market of financial instruments***.
- It focuses on the ***use of financial instruments*** (e.g., futures, options and swaps as well as credit derivatives) ***in terms of pricing, trading and risk management*** (e.g., hedging).
- Also, this part provides solid theoretical foundations of these products in order to expand the understanding of the role of financial instruments (e.g. derivatives) in market efficiency and financial crises.



Learning Outcomes

- C Evaluate the use of standardized derivatives for various purposes, including pricing and trading.
- D Apply financial instruments to hedge risk (including credit/market risk).

Lectures

- Lect. 7: Hedging Strategies Using Futures
 - Forward / Futures Price:
 - Stock, index, and currency futures
 - Hedging using forward and futures
- Lect. 8: Hedge equity portfolios
 - Basis Risk
 - Timing & asset mismatching
 - Cross-Hedging
 - Minimum variance hedging
 - Market risk & CAMP, changing Beta

Lectures

- Lect. 9: Swaps
 - Interest rate, currency, and index swaps
 - Comparative advantages
 - Design and value of swaps
 - Credit and liquidity risk

- Lect. 10: Options and trading strategies
 - Put-Call Parity and Capital Structure
 - Early Exercise of American Options
 - Strategies:
 - Naked, Covered, Bull, Bear, Butterfly

Lectures

- Lect. 11: Price options
 - Binomial Trees
 - No-arbitrage Pricing and Risk Neutral Pricing
 - Multiple-step Trees and American Options
 - Black-Scholes Option Price
- Lect. 12: Portfolio insurance and hedging option risks
 - Portfolio Insurance
 - Greek Letters and Hedging Options

Combinations of futures, swaps, and options and their advantages/disadvantages

- The combinations of futures, swaps, and options can provide some useful tools for hedging and risk management
- Advantages and disadvantages
 - Forward and Futures
 - Futures and index futures
 - Futures and Index options
 - An **index put option** can be used to hedge the market risk when market drops, like an insurance to your portfolio
 - Compared to the **index futures** strategy, there is a cost/premium for the index option, but you can enjoy a higher portfolio return when market moves up

The Exam: in general

- Covers **all six lecture** topics.
- In general
 - Understand the key concepts
 - Analysing and identifying risk types (e.g., margin trading, basis risk, market risk, interest rate risk, default risk, credit risk, liquidity risk, etc)
 - Pricing and hedging using forwards, futures, swaps, and options with different underlying assets (stock, index, currency, bond, commodity)

The Exam: in particular

- In particular,
 - **Portfolio hedging** use index, *index futures*.
 - **Portfolio insurance** using *index put options*
 - Interest and currency **swaps**
 - Option value and properties---time-value and intrinsic value, **put-call parity**
 - Option *trading strategies* (e.g. naked, covered, spreads)
 - **Option pricing**
 - *binomial trees: no-arbitrage and risk-neutral arguments*
 - Black-Scholes, European and American options (early exercise)
 - **Option portfolio hedging**: Greeks, delta, gamma, and vega neutral

Exam Questions, Formula, and Tables

- There will be **THREE** questions
- Each question is **20 marks**, containing calculations and general discussions on the implications and effectiveness.
- The same **formula sheet** and **tables** are provided in the exam.

The Exam

- Prepare your exam by going through the examples in the lecture slides and tutorial problems, and **make sure you understand how they work.**
- **Focus on economic motivations and intuitions,** not necessarily the details of the derivations.

Exam and Consultation

Exam Date:

10:00-12:00

Thursday, 11 June 2026

Consultation:

Office Hours:

Tuesday (9/6) and Wed (10/6)

15:00-17:00

Room: BS583

Email



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Student Module Feedback Questionnaire (MQ)

Please complete the Student **Module Feedback Questionnaire (MQ)** for this module before **Sunday, 31 May 2026**